
PROJECT PART II

420-635-AB-Network Installation and Administration I

OBJECTIVES

In this 1st part of the project, you will **configure a Virtual Host server**, and create websites identified by a combination of the following: **IP address, domain name and port number**.

GUIDELINES

The **Project Part 2** is graded out of **100** and accounts for **11%** of the final score.

You must provide a **detailed report** documenting the **commands executed in each section, the results obtained, and any relevant remarks. (Include clear and prominent screenshots.)**

You must also submit a **compressed file** containing of the **/etc/httpd/conf/httpd.conf**, **/etc/hosts** file, containing of **/var/www/html_project2** and **/var/www/virtuals** directories and the script.

The report must be submitted upon completion and before the deadline. Submission must be done via **Moodle**. The report file must be named: **FirstName_LastName_Report_P2.pdf**.

PONDERATION

Task	Graded on
Task 1 – Preparation	10%
Task 2 – Name-Based and Port-Based Virtual Hosts	20%
Task 3 – Name-Based Virtual Hosts with Access Control	20%
Task 4 – Virtual Web Servers by IP and Port	20%
Task 5 – Dynamic Virtual Hosting	15%
Report quality	15%
Total	100%

TASK 1 – PREPARATION (10 POINTS)

- The root directory for this project must be **"/var/www/html_project2"**.
- A web page named **master_project2.html** should be placed in this directory and configured as the default page for the root.
- The **master_project2.html** page must include hyperlinks allowing users to test each question and every scenario related to each question in this lab. See the example provided on the last page of this document.

TASK 2 – NAME-BASED AND PORT-BASED VIRTUAL HOSTS (20 POINTS)

- Create **virtual hosts** identified by both **hostname** and **port**, using the names provided below.
- All servers in this exercise will use the same IP address: **10.35.16.1**.
 - However, this address must be different from the main non-virtual website.
 - You will need to edit the **/etc/hosts** file accordingly to resolve the hostnames.
- Since these are virtual hosts, the basic Apache configuration will be similar for all of them:
 - Each server will have its **own directory** to store web pages.
 - Each server must also use its **own log files**.
- Virtual Hosts configuration details:
 - Hostname: **virtual1.aucegep.com**
 - Port: **80**
 - Document Root: **/var/www/virtuals/virtual1_80**
 - Hostname: **virtual2.aucegep.com**
 - Port: **80**
 - Document Root: **/var/www/virtuals/virtual2_80**
 - Hostname: **virtual1.aucegep.com**
 - Port: **8000**
 - Document Root: **/var/www/virtuals/virtual1_8000**
 - Hostname: **virtual2.aucegep.com**
 - Port: **8000**
 - Document Root: **/var/www/virtuals/virtual2_8000**

TASK 3 – NAME-BASED VIRTUAL HOSTS WITH ACCESS CONTROL (20 POINTS)

- Server IP addresses: **10.35.17.1**
- You are required to create virtual servers using **name-based hosting**, with **access permissions** and **document roots** as described below:
 - Hostname: **www.ici.com**
 - Port: **80**
 - Document Root: **/var/www/virtuals/ici**
 - Accessible to everyone (no restrictions).
 - Hostname: **intranet.ici.com**
 - Port: **80**
 - Document Root: **/var/www/virtuals/intranet**
 - Accessible to everyone (no restrictions).
 - Hostname: **intranet.ici.com**
 - Port **8000**
 - Document Root: **/var/www/virtuals/pre_production**
 - Accessible only from the **10.35.17.0/24** subnet.
 - Hostname: **development.ici.com**
 - Port **80**
 - Document Root: **/var/www/virtuals/development**
 - Accessible only from the **10.35.17.0/24** subnet.

TASK 4 – VIRTUAL WEB SERVERS BY IP AND PORT (20 POINTS)

- Create and configure **four virtual web servers** as described below.
- Use a separate directory for each server under:
/var/www/virtuals/q4/<server_name>
- For example, the **sales** web server will use:
/var/www/virtuals/q4/sales
- To access each server, use the IP address **10.35.16.1** followed by the corresponding port number. For example:
http://10.35.16.1:8080 should display the **sales** web server homepage.

- Virtual servers to create:
 - **Sales** server on port **8080**.
 - **Admin** server on port **8081**.
 - **Thing** server on port **8082**.
 - **Other** server on port **8083**.

TASK 5 – DYNAMIC VIRTUAL HOSTING (15 POINTS)

- Server IP address: **10.50.1.1**
- You are required to configure your Apache web server to support five virtual servers using only one **<VirtualHost>** block and the **VirtualDocumentRoot** directive.
- Virtual Servers to Configure:

Server Name	IP Address
www.itmt.com	10.50.1.1
www.itmt.ca	10.50.1.1
www2.itmt.com	10.50.1.1
www.montmo.com	10.50.1.1
www.montmo.ca	10.50.1.1

- Each website will have its own **directory** based on the following path format:

/var/www/virtuals/q5/*x/y/z*/

Where:

- **x** = TLD of the domain (com or ca).
- **y** = The name of the company.
- **z** = The prefix before the domain (e.g., www, www2).
- For example, **www.itmt.com**, the full directory would be:
/var/www/virtuals/q5/*com/itmt/www*/
- All virtual host errors should be logged to:
/var/www/virtuals/q5/logs/virtual_error_q5_log

Note: For this question you can use **tree** to help you visualize the directory structure.

Project Part 2

Task 2

<http://virtual1.aucegep.com/>

<http://virtual1.aucegep.com:8000/>

<http://virtual2.aucegep.com/>

<http://virtual2.aucegep.com:8000/>

Task 3

<http://www.ici.com>

<http://intranet.ici.com>

<http://intranet.ici.com:8000>

<http://development.ici.com>

Task 4

<http://10.35.16.1:8080>

<http://10.35.16.1:8081>

<http://10.35.16.1:8082>

<http://10.35.16.1:8083>

Task 5

<http://www.itmt.com>

<http://www.itmt.ca>

<http://www2.itmt.com>

<http://www.montmo.com>

<http://www.montmo.ca>